

Mohammed Raza

+91 9493631510 | mohammed.raza@research.iiit.ac.in | LinkedIn | GitHub

EDUCATION

IIIT Hyderabad <i>B.Tech. in Computer Science, M.S. by Research in Computer Science and Engineering</i> • DSA, Operating Systems, Computer Networks, Algorithms, Probability & Statistics, Real Analysis, Computer Architecture	CGPA : 6.08 / 10 Aug. 2024 – June 2029
FIITJEE Vizag <i>BIEAP(12th Board)</i>	Percentage : 97% Aug. 2022 – May 2024
Sri Chaitanya Techno School Vizag <i>CBSE(10th Board)</i>	Percentage : 96.2% Aug. 2019 – May 2022

PROJECTS

Multilingual Healthcare Communication Platform <i>Node.js, React, Supabase, PostgreSQL, Gemini API, Bhashini APIs</i> • Built a real-time multilingual hospital communication system using Bhashini STT, TTT, TTS, and S2S APIs • Implemented low-latency STT → Translation → TTS pipeline with Node.js backend and Supabase/PostgreSQL	GitHub
Research on Biconnected Components <i>Python, NetworkX, Matplotlib</i> • Benchmarked Tarjan, Vishkin–Slota, and Maduri algorithms on real-world graph datasets • Visualized and analyzed time-complexity trends using Matplotlib	GitHub
Network File System <i>C, Distributed Systems, Pthreads</i> • Implemented distributed file storage with concurrent reads and single-writer consistency • Used Pthread mutexes for synchronization and fault-safe access control	GitHub
C-Shell (Unix Shell) <i>C, Unix Systems Programming</i> • Built a Unix shell supporting piping, I/O redirection, signal handling, and job control • Implemented foreground/background process management and command parsing	GitHub

ACHIEVEMENTS & HONORS

- **Codeforces (Rating 1251):** Demonstrated proficiency in data structures, algorithms, and competitive programming through consistent performance. (Handle)
- **LeetCode:** Solved **300+** problems with consistent participation in weekly contests; specialized in optimization techniques. (Profile)
- **JEE Main 2024:** Secured **AIR 75000** among **1.2M+** candidates
- **JEE Advanced 2024:** Achieved **AIR 17000** in one of the world's most competitive entrance examinations
- **UGEE 2024:** Achieved **AIR 79**

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL, Bash, x86-64 Assembly.
Core Cs: Data Structures and Algorithms, Operating Systems, Computer Networks
Developer Tools: Git, Docker, VS Code, Visual Studio, Linux, Postman, Hugging Face, Reverse Engineering
Libraries: pandas, NumPy, Matplotlib, PyTorch
IOT: Sensor Data Processing, MQTT, Real-time Data Visualization.
Web & Backend: React, Node.js, REST APIs, MongoDB, Firebase, Supabase.